User guide

How to start GreenLCC India, what each screen does, and how to read the results.

1. Start the tool

The tool runs on your own computer, in your web browser, at an address starting with **http://localhost**. Nothing is uploaded anywhere.

System	Steps
Windows	Install Python 3.10 or newer (tick "Add python.exe to PATH"). Unzip the folder to a short path such as **C:\GreenLCC**. Double-click **Start_GreenLCC.bat**. The first run sets up a private environment (3-5 minutes, internet needed); later runs start in seconds. Your browser opens the app. Keep the black window open; close it to stop the tool.
macOS / Linux	In a terminal in the folder: **sh start_greenlcc.sh**, then open the address it prints.
Any system, by hand	**python -m venv .venv**, activate it, **pip install -r requirements.txt**, then **python launcher.py**.

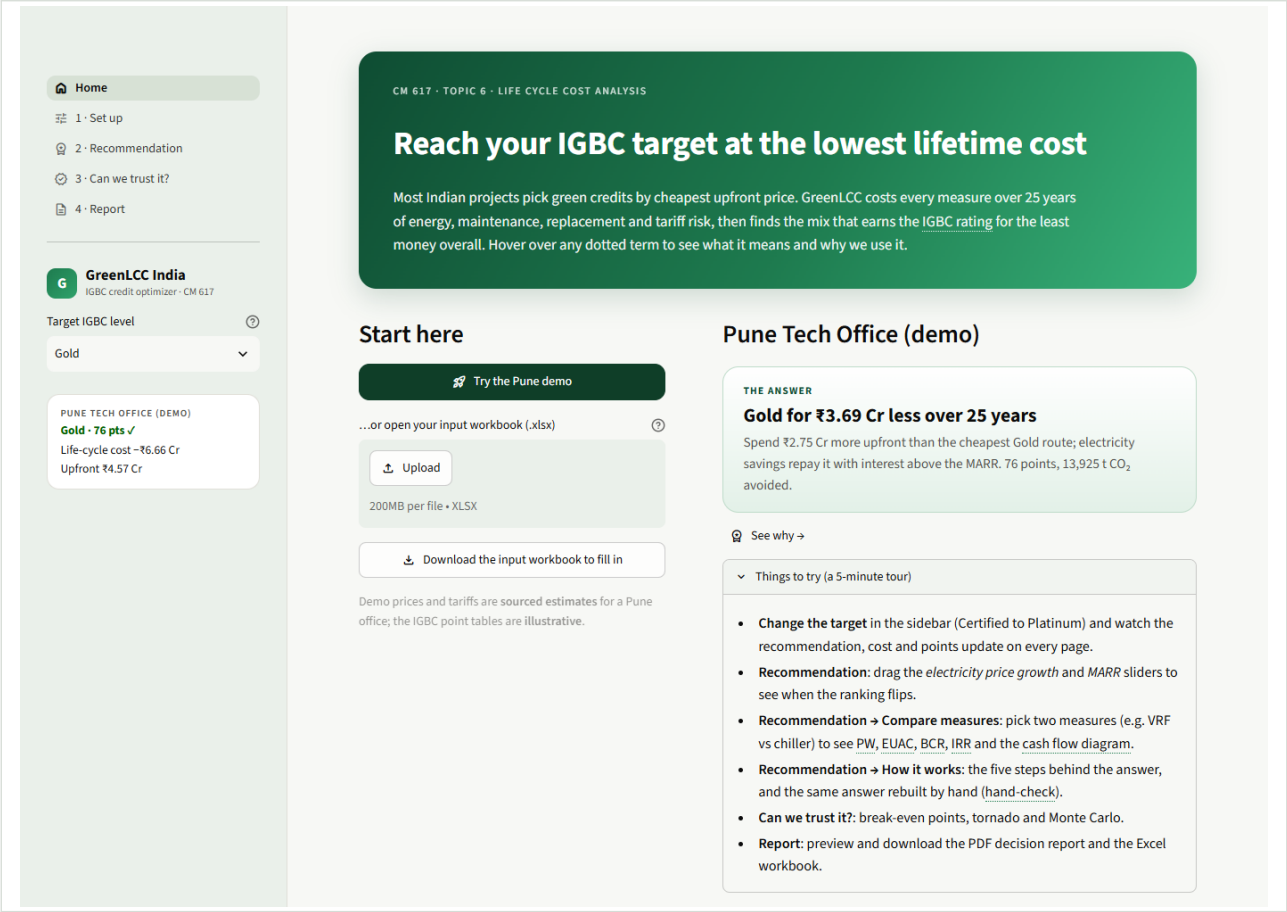
If you close the browser tab, double-click **Open GreenLCC in browser.url** while the black window is still open.

2. The five screens

A menu on the left lists the screens in order. Each page ends with a **Next** button. The target IGBC level is chosen in the sidebar and applies everywhere.

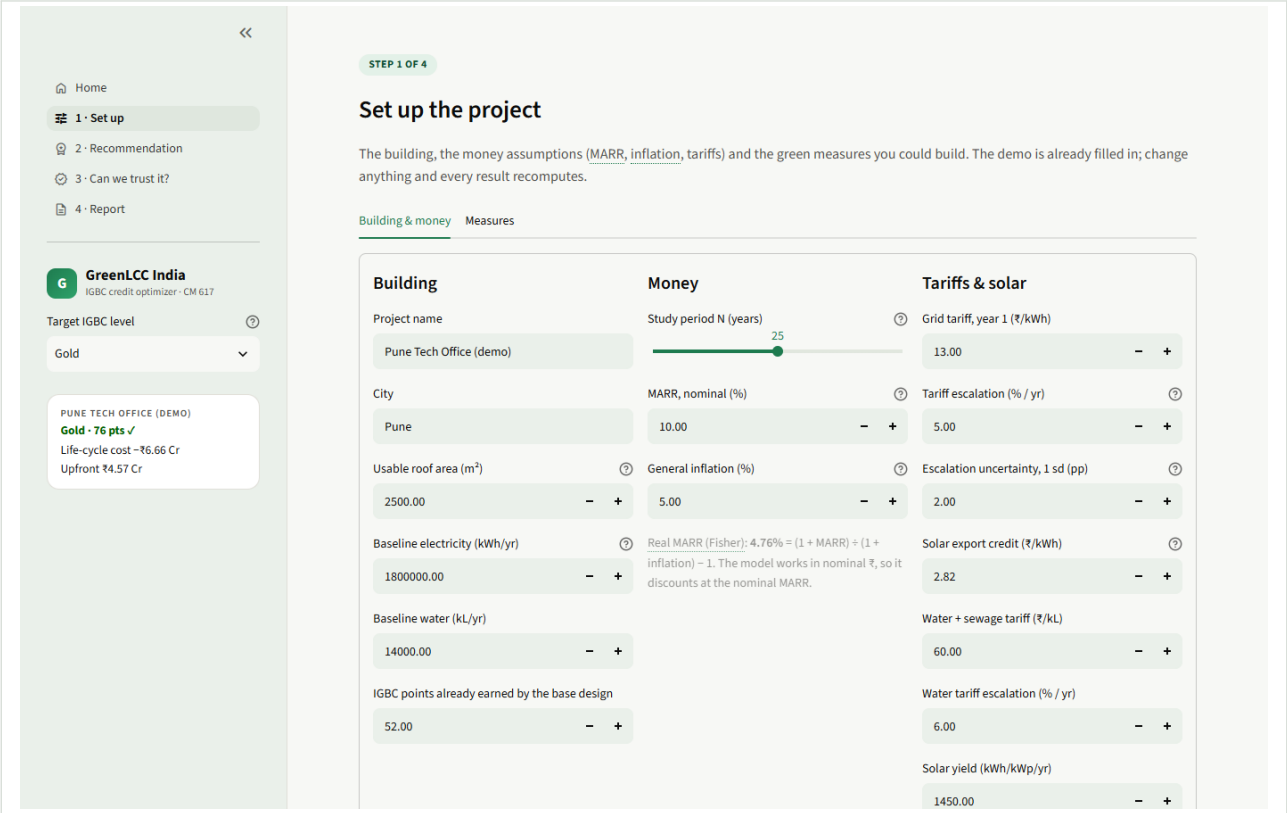
Home

Click **Try the Pune demo** to load the demo building, or open your own input workbook. The page also shows what the input workbook contains, sheet by sheet, and a short "Things to try" list.



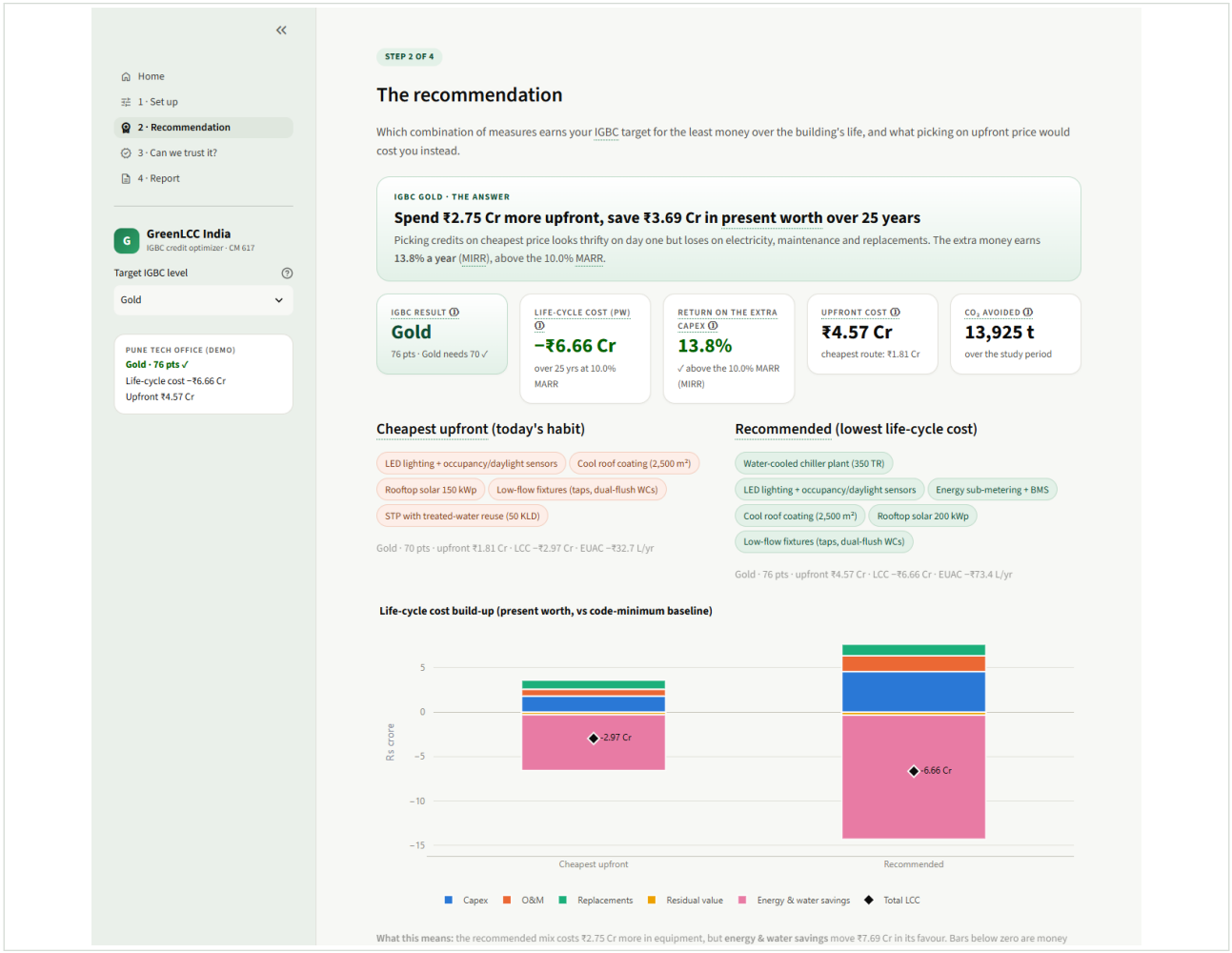
Step 1: Set up

Check or change the building, the money assumptions (required return, inflation, tariffs, solar) and the list of green measures. Press **Apply changes**; every result recomputes.

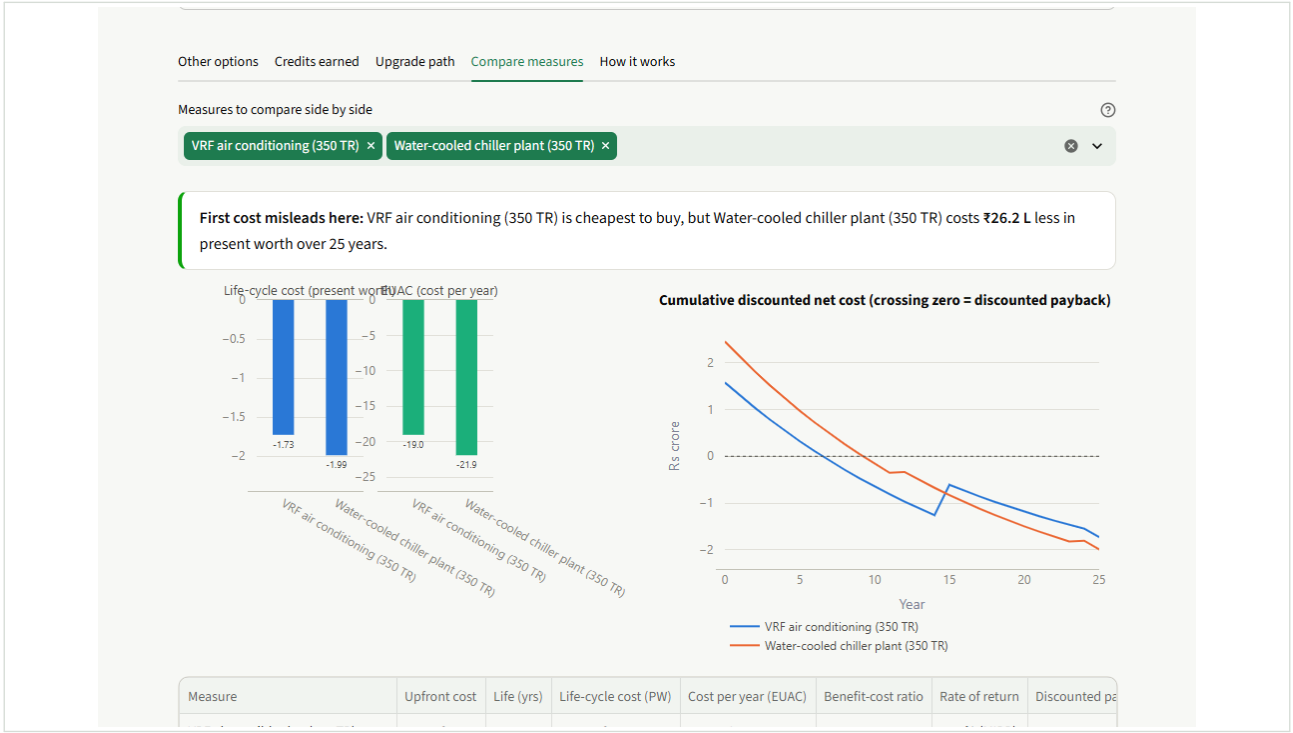


Step 2: Recommendation

The answer in one sentence, then the key figures, the cheapest-upfront route beside the recommended route, a cost build-up chart and a quick what-if with two sliders. Tabs below show other options, credits earned, the upgrade path from level to level, a side-by-side comparison of measures and the “How it works” explanation.

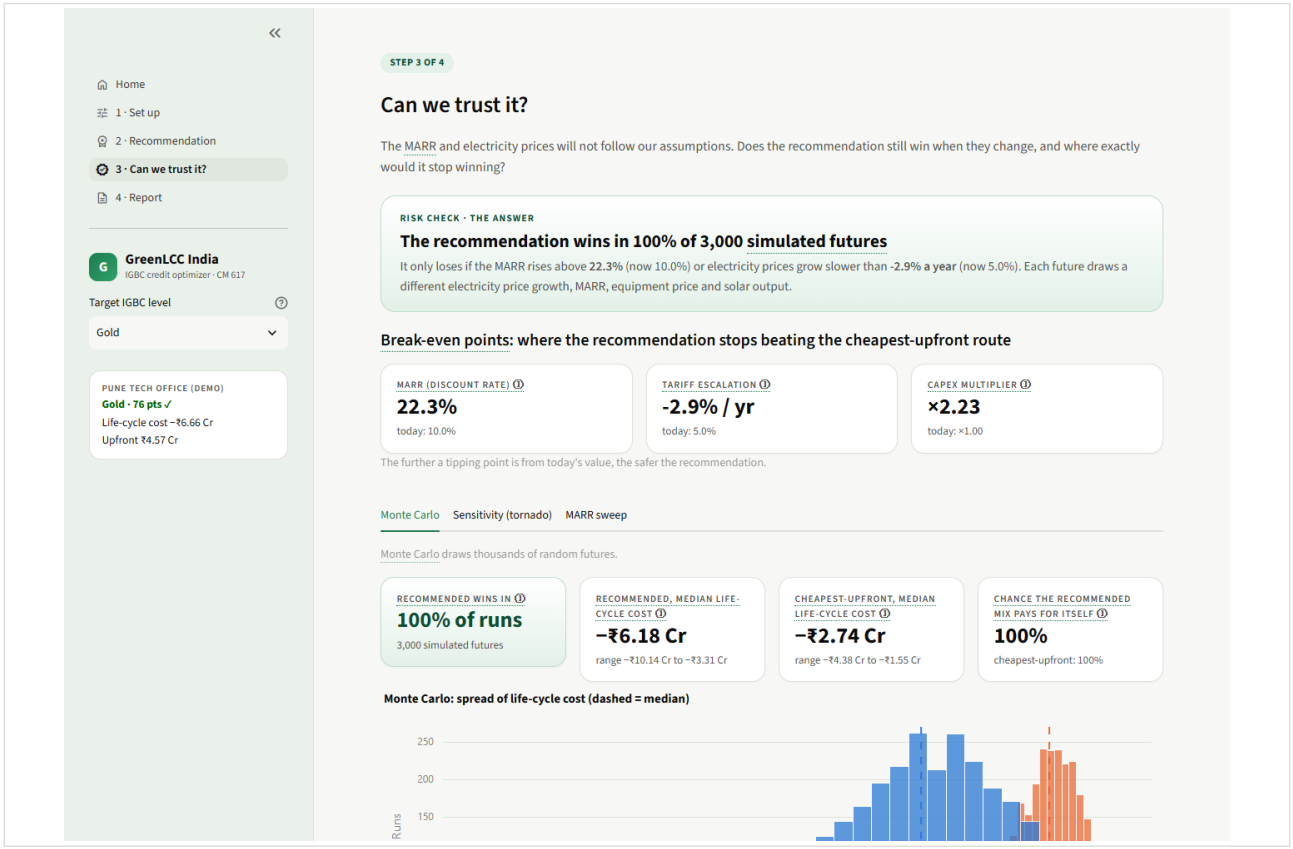


Compare measures: pick up to five measures to see life-cycle cost, cost per year, benefit-cost ratio, rate of return, discounted payback and a cash flow diagram.



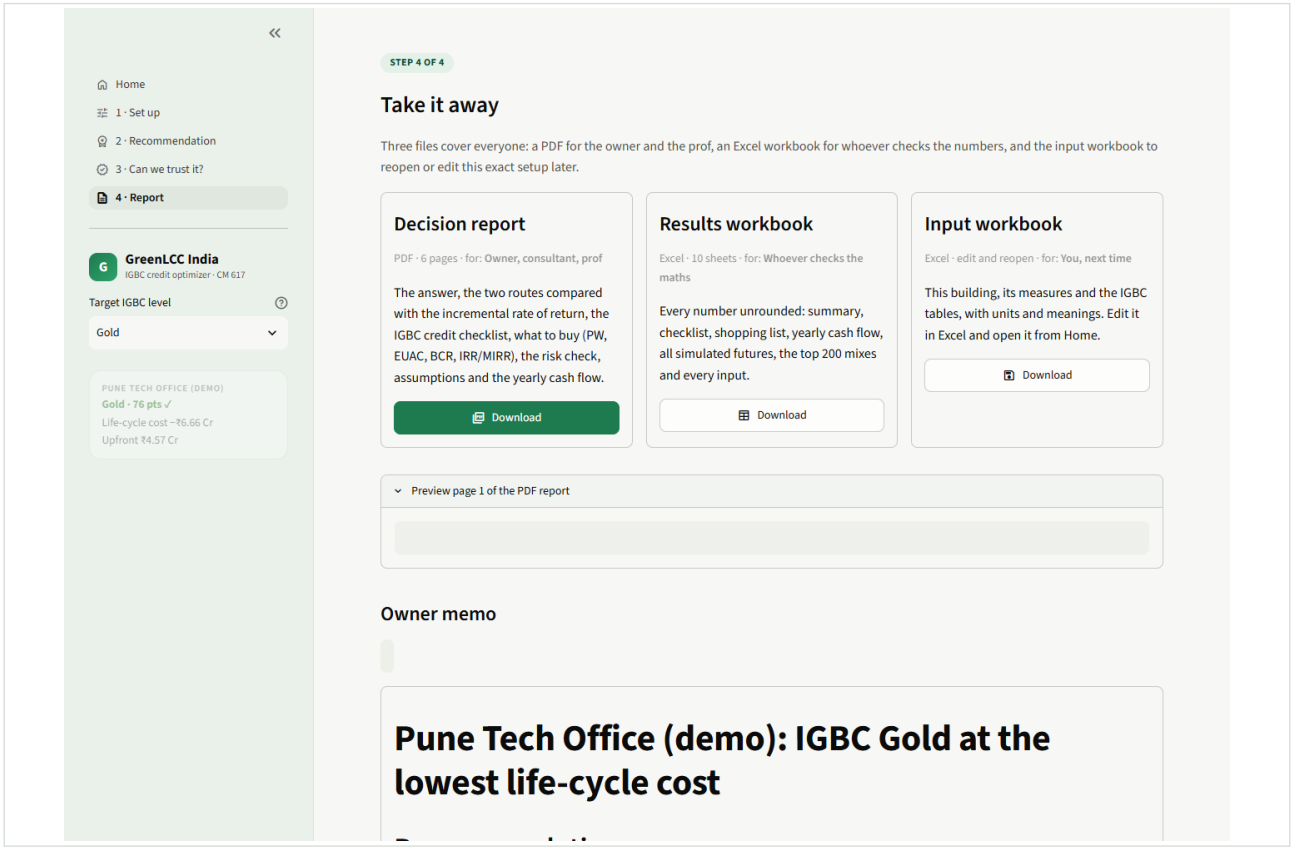
Step 3: Can we trust it?

Shows where the recommendation would stop winning (break-even points), how often it wins across thousands of random futures (Monte Carlo), which assumptions matter most (tornado chart) and how it changes with the required return (MARR sweep).



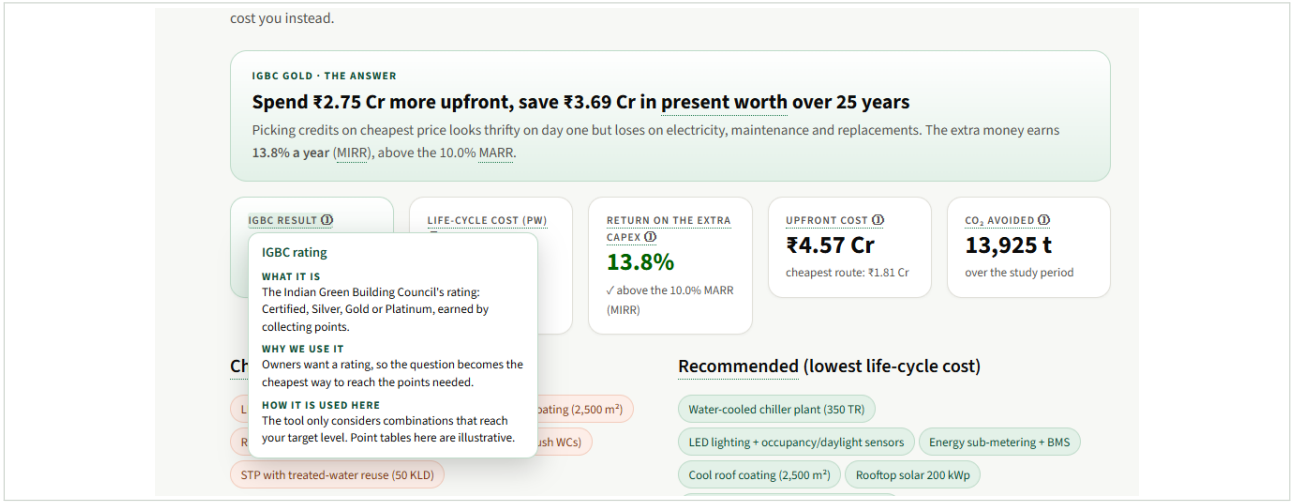
Step 4: Report

Download a plain-language PDF decision report for the owner, a results workbook with every number, and the input workbook to reopen later. A preview of the PDF is shown on the page, and the owner memo can be copied.



3. Hover over dotted words

Any word with a dotted underline (MARR, present worth, EUAC, Monte Carlo and more) opens a popup when you hover over it, or tab to it, or tap it. The popup says what the term is, why the tool uses it and how it is used here. The small **?** next to input boxes gives the same kind of help.



4. Use your own project

- On Home, click **Download the input workbook to fill in**. Open it in Excel.
- Read the README sheet; then overwrite the values in the **Project** sheet (type percentages as numbers, for example 10 for 10%), and edit the **Measures** sheet.
- Costs and savings are **incremental**: how much more, or how much less, than the ordinary (code-minimum) option. A VRF row holds the VRF price minus the split air-conditioners it replaces.

- Rows that share a **Group** are alternatives (for example one HVAC system or one solar size); only one of them is chosen.
- Leave the part columns blank if a measure has no separately replaced part.
- Save the file, then open it on Home. Any error is listed with its sheet and row.

5. Reading the results

Number	Good when	Meaning
Life-cycle cost (present worth)	Lowest; negative is best	All costs minus savings over 25 years, in today's rupees. This decides the ranking.
Cost per year (EUAC)	Lowest	The same cost spread evenly per year. Ranks exactly like present worth.
Benefit-cost ratio	Above 1	Savings are worth more than the cost, at the MARR.
Rate of return (IRR / MIRR)	Above the MARR	Use it to accept or reject, not to rank.
Return on the extra capex	At least the MARR	Whether the costlier route's extra money is well spent. The headline.
Break-even	Far from today's value	How wrong an assumption can be before the answer changes.
Win rate (Monte Carlo)	Close to 100%	Share of simulated futures where the recommendation wins.

6. If something goes wrong

Problem	Fix
"python is not recognised"	Reinstall Python and tick "Add python.exe to PATH".
Browser did not open	Go to the address printed in the black window, for example http://localhost:8501.
Port already in use	The tool picks the next free port automatically and updates the .url shortcut.
Install error about a missing file or a long path	Move the folder to a shorter path such as C:\GreenLCC, delete the .venv folder and run again.
A page looks blank	Wait a few seconds; pages take about two seconds to load. Refresh if it stays blank.
Want to start over	On Home click "Clear the project and start over".

Optional AI: the only AI feature in the tool polishes the owner memo wording on the Report page. It needs **pip install -r requirements-ai.txt** and an ANTHROPIC_API_KEY. Without it a built-in template is used and everything works offline.